

Lecture 9

Condition Variables and Semaphores

Previously ...

- We have seen how **locks** may be used to execute critical sections of code with mutual exclusion, so avoiding race conditions
- We have seen how **locks** may be **implemented** using:
 - Some hardware support (in the form of a more powerful instruction)
 - Some operating system support and a queue

CONDITION VARIABLES

Threads Synchronization

- Lock is not the only primitive needed in concurrent programs
- There are many cases where a thread wishes to check whether a condition is true before continuing its execution
- A parent thread might check whether a child thread has completed before continuing
- There are 2 questions
 - How does a child indicate that it is done?
 - How does a parent wait for a child?

Spinning until 'true'

```
1  volatile int done = 0;
2
3  void *
4  child(void *arg) {
5      printf("child\n");
6      done = 1;
7      return NULL;
8  }
9
10 int
11 main(int argc, char *argv[]) {
12     printf("parent: begin\n");
13     pthread_t c;
14     Pthread_create(&c, NULL, child, NULL); // create child
15     while (done == 0)
16         ; // spin
17     printf("parent: end\n");
18     return 0;
19 }
```

Desired output

```
parent: begin
child
parent: end
```

inefficient as the parent spins
and wastes CPU time

Condition Variable

- A condition variable is an explicit queue that threads can put themselves on when some state of execution is not as desired (Waiting on the condition)
- Some other thread, when it changes the said state, can then wake the waiting threads allowing them to continue (Signaling on the condition)

```
pthread_cond_wait(pthread_cond_t *c, pthread_mutex_t *m);  
pthread_cond_signal(pthread_cond_t *c);
```

Example

```
pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;
pthread_cond_t  init = PTHREAD_COND_INITIALIZER;

Pthread_mutex_lock(&lock);
while (initialized == 0)
    Pthread_cond_wait(&init, &lock);
Pthread_mutex_unlock(&lock);
```

```
Pthread_mutex_lock(&lock);
initialized = 1;
Pthread_cond_signal(&init);
Pthread_mutex_unlock(&lock);
```

- A thread checks to see if the variable `initialized` has yet been set to something other than zero
- If not, the thread simply calls the wait routine in order to sleep until some other thread wakes it
- The wait call, in addition to putting the calling thread to sleep, releases the lock
- When signaling (as well as when modifying the global variable `initialized`), we always make sure that we hold the lock

Wait ()

- `wait()` takes a mutex as a parameter; it assumes that this mutex is locked when `wait()` is called.
- `wait()` will
 - Release the lock and put the calling thread to sleep
 - When the thread wakes up (after some other thread has signaled it) ...
 - it must re-acquire the lock before returning to the caller

```
1  int done = 0;
2  pthread_mutex_t m = PTHREAD_MUTEX_INITIALIZER;
3  pthread_cond_t c = PTHREAD_COND_INITIALIZER;
4
5  void thr_exit() {
6      Pthread_mutex_lock(&m);
7      done = 1;
8      Pthread_cond_signal(&c);
9      mutex_unlock(&m);
10 }
11
12 void *child(void *arg) {
13     printf("child\n");
14     thr_exit();
15     return NULL;
16 }
17
18 void thr_join() {
19     Pthread_mutex_lock(&m);
20     while (done == 0)
21         Pthread_cond_wait(&c, &m);
22     Pthread_mutex_unlock(&m);
23 }
24
25 int main(int argc, char *argv[]) {
26     printf("parent: begin\n");
27     pthread_t p;
28     Pthread_create(&p, NULL, child, NULL);
29     thr_join();
```

**CODING TIP: ALWAYS HOLD
THE LOCK WHILE SIGNALING**



Producer/Consumer (Bounded Buffer)

- One or more Producer threads and one or more Consumer threads
 - Producers produce data items and wish to put them in a buffer
 - Consumers get data items out of the buffer and consume the data in some way
- Examples:
 - A multi-threaded web server – a producer puts requests into the buffer and consumer threads get and process them
 - Unix pipe: `grep foo file.txt | wc -l` (grep is the producer and wc the consumer)
- The bounded buffer is a shared resource and requires synchronized access to it
- Race conditions must be prevented

Buffer (A Single int)

```
1  int buffer;
2  int count = 0; // initially, empty
3
4  void put(int value) {
5      assert(count == 0);
6      count = 1;
7      buffer = value;
8  }
9
10 int get() {
11     assert(count == 1);
12     count = 0;
13     return buffer;
14 }
```

Assume buffer is empty

Mark it as 'full'

Take the data and make the buffer empty again

Producer and Consumer Code

```
1 void *producer(void *arg) {
2     int i;
3     int loops = (int) arg;
4     for (i = 0; i < loops; i++) {
5         put(i);
6     }
7 }
8
9 void *consumer(void *arg) {
10     int i;
11     while (1) {
12         int tmp = get();
13         printf("%d\n", tmp);
14     }
15 }
```

puts an integer into the shared buffer
'loops' number of times

gets the data out of that shared
buffer (forever)

Notes

- The `put()` and `get()` functions have critical sections within them
 - `put()` updates the buffer
 - `get()` reads from it
 - e.g. two producers might try to update the buffer at the same time and an item could be lost
- But putting a lock around the critical section alone does not work
- We also need condition variables
- Let's see how ...

```

1  cond_t  cond;
2  mutex_t mutex;
3
4  void *producer(void *arg) {
5      int i;
6      for (i = 0; i < loops; i++) {
7          mutex_lock(&mutex);
8          while (count == 1)
9              cond_wait(&cond, &mutex);
10         put(i);
11         cond_signal(&cond);
12         mutex_unlock(&mutex);
13     }
14 }
15
16 void *consumer(void *arg) {
17     int i;
18     for (i = 0; i < loops; i++) {
19         mutex_lock(&mutex);
20         while (count == 0)
21             cond_wait(&cond, &mutex);
22         int tmp = get();
23         cond_signal(&cond);
24         mutex_unlock(&mutex);
25         printf("%d\n", tmp);
26     }
27 }

```

Scenario with
two consumers (Tc1 and Tc2)
and a producer (Tp)

If Tc1 and Tc2 run first, what
might be a possible sequence
of events?

Problem

- What is the problem with this code?
 - Suppose the two consumers run first (Tc1 and Tc2), and both go to sleep (line 21) as the buffer is empty
 - Then, the producer runs, puts a value in the buffer, wakes one of the consumers (say Tc1), and goes to sleep
 - Tc1 then runs (returning from wait() at line 21), re-checks the condition (line 20), finds the buffer is full, and gets the value (line 22)
 - This consumer then signals on the condition, waking one of the threads that is sleeping
 - However, which thread should be woken?

```

1  cond_t  empty, fill;
2  mutex_t mutex;
3
4  void *producer(void *arg) {
5      int i;
6      for (i = 0; i < loops; i++) {
7          mutex_lock(&mutex);
8          while (count == 1)
9              cond_wait(&empty, &mutex);
10         put(i);
11         cond_signal(&fill);
12         mutex_unlock(&mutex);
13     }
14 }
15
16 void *consumer(void *arg) {
17     int i;
18     for (i = 0; i < loops; i++) {
19         mutex_lock(&mutex);
20         while (count == 0)
21             cond_wait(&fill, &mutex);
22         int tmp = get();
23         cond_signal(&empty);
24         mutex_unlock(&mutex);
25         printf("%d\n", tmp);
26     }
27 }

```

Solution

Use two condition variables in order to properly signal which type of thread should wake up when the state of the system changes

Use While (not If)

- Why do we use **while** and not **if** to check the condition?
 - Suppose we have two consumers (Tc1 and Tc2) as before
 - Tc1 runs and goes to sleep (line 21) as the buffer is empty
 - Tc2 is ready, but does not yet run
 - Then, the producer runs, puts a value in the buffer, wakes Tc1, and goes to sleep
 - Before Tc1 returns from the wait() and reacquires the lock, Tc2 runs, finds an item in the buffer and consumes it
 - Then Tc1 returns from the wait() at line 21 – using **if** would allow it to get an item from an empty buffer

Using while is always correct – using if does not always work

Final Put and Get

```
1  int buffer[MAX];
2  int fill  = 0;
3  int use   = 0;
4  int count = 0;
5
6  void put(int value) {
7      buffer[fill] = value;
8      fill = (fill + 1) % MAX;
9      count++;
10 }
11
12 int get() {
13     int tmp = buffer[use];
14     use = (use + 1) % MAX;
15     count--;
16     return tmp;
17 }
```

Circular Buffer

The single int has been extended into a *circular* buffer of size MAX

- fill is the input pointer
- use is the output pointer

Final Producer/ Consumer

```
1  cond_t empty, fill;
2  mutex_t mutex;
3
4  void *producer(void *arg) {
5      int i;
6      for (i = 0; i < loops; i++) {
7          mutex_lock(&mutex);
8          while (count == MAX)
9              cond_wait(&empty, &mutex);
10         put(i);
11         cond_signal(&fill);
12         mutex_unlock(&mutex);
13     }
14 }
15
16 void *consumer(void *arg) {
17     int i;
18     for (i = 0; i < loops; i++) {
19         mutex_lock(&mutex);
20         while (count == 0)
21             cond_wait(&fill, &mutex);
22         int tmp = get();
23         cond_signal(&empty);
24         mutex_unlock(&mutex);
25         printf("%d\n", tmp);
26     }
27 }
```

SEMAPHORES

Introduction

- We have seen that we need both locks and **condition variables** to solve a broad range of relevant and interesting concurrency problems
 - **Locks** can be used to ensure **mutual exclusion**
 - **Condition variables** can be used for **synchronize**
- Now ... we can actually use semaphores as both locks and condition variables

Definition

- A semaphore is as an object with an integer value that we can manipulate with two routines
- In the POSIX standard, these are called
 - `sem_wait()`
 - `sem_post()`
- The initial value of the semaphore determines its behavior
 - `sem_init()` is also needed

How it Works

```
#include <semaphore.h>
sem_t s;
sem_init(&s, 0, 1);
```

0 indicates semaphore shared
between threads in same process

s is initialized to a value of 1

Returns immediately or causes the
caller to wait for a subsequent post

```
int sem_wait(sem_t *s) {
    wait until value of semaphore s is greater than 0
    decrement the value of semaphore s by 1
}
```

Does not ever suspend the
caller, just wakes a waiting thread

```
int sem_post(sem_t *s) {
    increment the value of semaphore s by 1
    if there are 1 or more threads waiting, wake 1
}
```

Binary Semaphores

- Binary Semaphore can be used as a lock

```
sem_t m;  
sem_init(&m, 0, X); // initialize semaphore to X; what should X be?  
  
sem_wait(&m);  
// critical section here  
sem_post(&m);
```

- Two threads are trying to enter their critical sections
 - Suppose one thread holds the lock, i.e. it has called sem wait(), but not yet called sem post()
 - If another thread tries to enter the critical section it will find that the value of the semaphore is 0, and thus wait

Semaphore as Condition Variable

- Semaphores are also useful when a thread wants to halt its own progress waiting for something to change

```
void *
child(void *arg) {
    printf("child\n");
    // signal here: child is done
    return NULL;
}

int
main(int argc, char *argv[]) {
    printf("parent: begin\n");
    pthread_t c;
    Pthread_create(c, NULL, child, NULL);
    // wait here for child
    printf("parent: end\n");
    return 0;
}
```

What should the initial value of this semaphore be?

```
sem_t s;

void *
child(void *arg) {
    printf("child\n");
    // signal here: child is done
    sem_post(&s);
    return NULL;
}

int
main(int argc, char *argv[]) {
    sem_init(&s, 0, X); // what should X be?
    printf("parent: begin\n");
    pthread_t c;
    Pthread_create(c, NULL, child, NULL);
    // wait here for child
    sem_wait(&s);
    printf("parent: end\n");
    return 0;
}
```

Parent waits for its child to complete

Producer / Consumer Problem

- Introduces two semaphores, *empty* and *full*

```
void *producer(void *arg) {
    int i;
    for (i = 0; i < loops; i++) {
        sem_wait(&empty);
        put(i);
        sem_post(&full);
    }
}
```

Producer waits for a buffer slot to become empty in order to put data into it
put(i) and get() are done without locks
Race condition may occur

```
void *consumer(void *arg) {
    int i, tmp = 0;
    while (tmp != -1) {
        sem_wait(&full);
        tmp = get();
        sem_post(&empty);
        printf("%d\n", tmp);
    }
}
```

Consumer similarly waits for a buffer slot to become filled before using it

// line c1
// line c2
// line c3

```
int main(int argc, char *argv[]) {
    // ...
    sem_init(&empty, 0, MAX); // MAX buffers are empty to begin with.
    sem_init(&full, 0, 0);    // ... and 0 are full
    // ...
}
```

How can locks be added ?

With Locks

```
void *producer(void *arg) {
    int i;
    for (i = 0; i < loops; i++) {
        sem_wait(&mutex);
        sem_wait(&empty);
        put(i);
        sem_post(&full);
        sem_post(&mutex);
    }
}
```

Think of a scenario that may cause deadlock from this code

```
void *consumer(void *arg) {
    int i;
    for (i = 0; i < loops; i++) {
        sem_wait(&mutex);
        sem_wait(&full);
        int tmp = get();
        sem_post(&empty);
        sem_post(&mutex);
        printf("%d\n", tmp);
    }
}
```

How do we fix it ?

```
void *producer(void *arg) {  
    int i;  
    for (i = 0; i < loops; i++) {  
        sem_wait(&empty);  
        sem_wait(&mutex);  
        put(i);  
        sem_post(&mutex);  
        sem_post(&full);  
    }  
}
```

Solution – Reduce the Scope of the Lock

```
void *consumer(void *arg) {  
    int i;  
    for (i = 0; i < loops; i++) {  
        sem_wait(&full);  
        sem_wait(&mutex);  
        int tmp = get();  
        sem_post(&mutex);  
        sem_post(&empty);  
        printf("%d\n", tmp);  
    }  
}
```

How to Implement a Semaphore

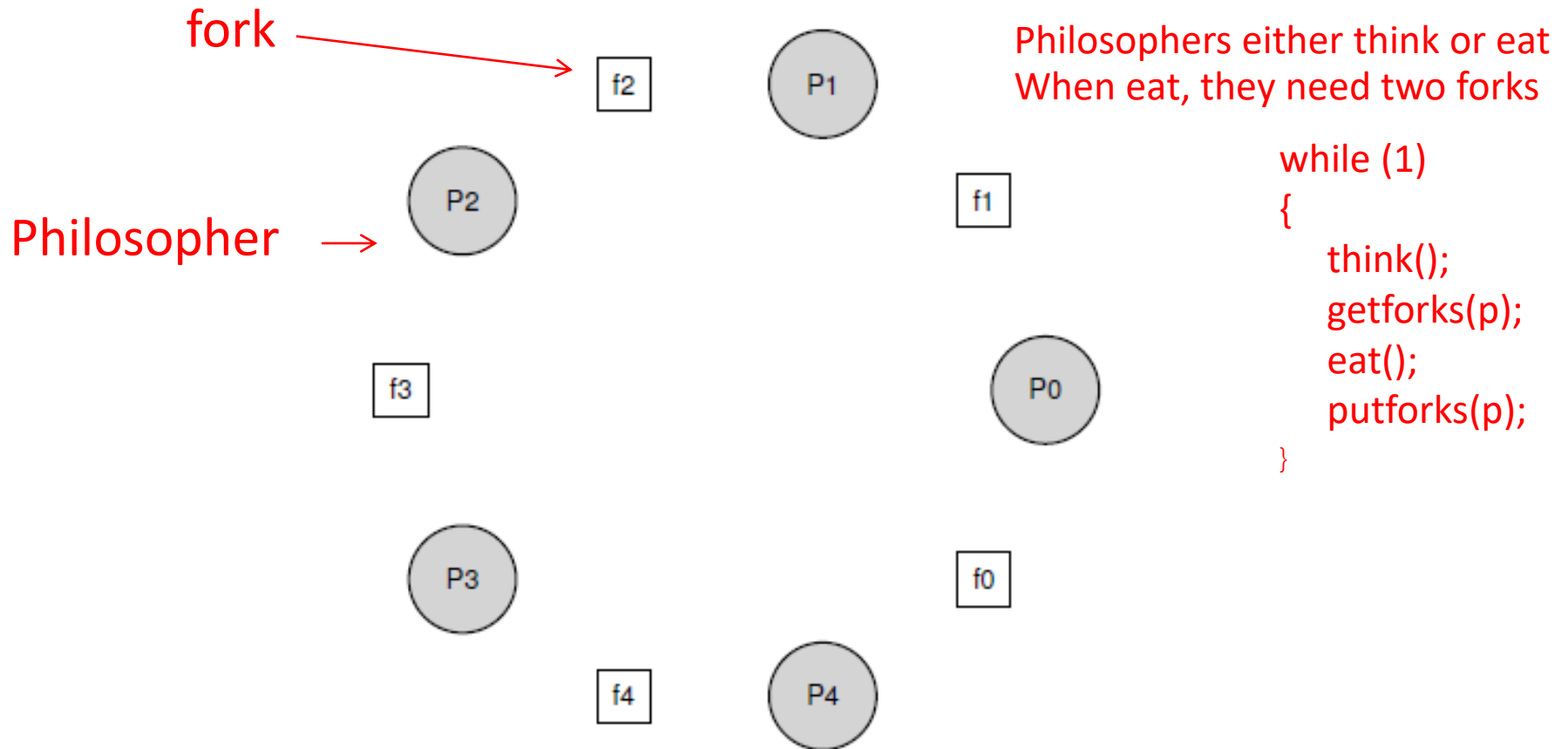
```
typedef struct __Sem_t {
    int value;
    pthread_cond_t cond;
    pthread_mutex_t lock;
} Sem_t;

// only one thread can call this
void Sem_init(Sem_t *s, int value) {
    s->value = value;
    Cond_init(&s->cond);
    Mutex_init(&s->lock);
}
```

```
void Sem_wait(Sem_t *s) {
    Mutex_lock(&s->lock);
    while (s->value <= 0)
        Cond_wait(&s->cond, &s->lock);
    s->value--;
    Mutex_unlock(&s->lock);
}

void Sem_post(Sem_t *s) {
    Mutex_lock(&s->lock);
    s->value++;
    Cond_signal(&s->cond);
    Mutex_unlock(&s->lock);
}
```

The Dining Philosophers



Write the routines `getforks(p)` and `putforks(p)` such that there is no deadlock, no philosopher starves, and as many philosophers can eat at the same time as possible

Some Hints

- When philosopher p wishes to refer to the fork on their left, they simply call `left(p)`
- Similarly, the fork on the right of a philosopher is referred to by calling `right(p)`
 - `int left(int p) { return p; }`
 - `int right(int p) { return (p + 1) % 5; }`
- Declare the semaphores as follows:
 - `sem_t forks[5];`

Now do it